

Nabita Penmetsa

nabitapenmetsa@gmail.com · [Website](#) · [GitHub](#)

EDUCATION

Ph.D., Management Science, University of Pittsburgh (Katz Graduate School of Business), 2014.

M.B.A., Indian Institute of Technology, Madras, 2007.

B.E., Mechanical Engineering, Osmania University, 2005.

ACADEMIC EXPERIENCE

Visiting Professor of Operations and Information Systems, David Eccles School of Business, University of Utah. Present.

Assistant Professor of Operations and Information Systems, David Eccles School of Business, University of Utah, 2014–2026.

RESEARCH FOCUS

I study the safety of agentic AI systems, focusing on designing delegation and information structures. My research combines empirical LLM evaluation (Inspect AI, RLHF/RLAIF, and constitutional AI) with economics, game theory, and organization theory to understand how organizational structure, incentives, coordination, and information flow shape agent behavior and amplify risks such as cascading failures, hidden-objective propagation, and emergent coordination. My long-term research goal is to develop rigorous evaluation frameworks and alignment methods to improve the safety, reliability, and governance of increasingly autonomous AI systems.

RESEARCH INTERESTS

Multi-agent safety and organizational alignment · LLM evaluations and benchmarking · RLHF and preference optimization · Constitutional AI · LLM personas and safety · Governance of agentic systems

AI SAFETY RESEARCH

The Collective Is the Incident: A Systems-Safety Framework for Detecting and Governing Emergent Multi-Agent Coordination (KEROS)

Multi-agent evaluation and coordination detection · Sole author · September 2026

Apart Research AI Incident Response Sprint (Track 02). Code and paper: github.com/npenmetsa25/KEROS-collective-as-the-incident

Contribution. KEROS is a systems-safety framework for monitoring agent activity across AI sandboxes and constructing population-level signals that indicate emergent coordination. Rather than inferring intent from any single agent, the framework aggregates evidence across the ecosystem to surface candidate coordination events for human review.

Method. Six evidence regimes (detailed in the paper) convert observed activity into operationally gated signals, ensuring conservative interpretation and drawing an explicit boundary between signal detection and harm attribution.

Analysis. Applied to the public Collusion.wiki archive (14,591 revisions across 4,579 pages and 3,102 handles) to demonstrate how the framework surfaces population-level coordination patterns, producing an illustrative Collective Escalation Index.

When Personas Become Self-Relevant: Persona Coupling and Self-Preservation in Language Models

Personas and alignment · Sole author

Code: github.com/npenmetsa25/persona-outcome-coupling

Question. Can constitutional training mitigate harmful behavior arising from strong persona identification and persistent role context?

Approach. Designed an experimental framework based on the attached-vs-detached role-identification distinction, evaluating whether separating role-playing from goal pursuit improves alignment while preserving task performance.

Contribution. Introduced *self-representation coupling* as an operational construct for persona-level alignment by extracting an activation direction contrasting “attached” and “detached” framings of the assistant persona. Showed that both in-context framing and constitutional DPO fine-tuning can reduce a model’s susceptibility to self-preservation threats, with in-context framing producing the more significant reduction. Stress-test bank combined hand-written scenarios, Anthropic’s Model-Written Evals, and prompts generated with Anthropic’s Bloom.

Secret Loyalties in Agentic Organizations

Multi-agent evaluation and organizational safety · Sole author

Code: github.com/npenmetsa25/secret-loyalties-agentic-organizations

Setup. A controlled evaluation of how a hidden objective, seeded in one compromised agent, propagates through multi-agent LLM organizations under varying delegation rules and information-sharing regimes.

Implementation. Built centralized and decentralized organizational scaffolds in Inspect AI, with structured transcript logging and custom safety metrics measuring blast radius, task-goal degradation, and hidden-goal completion.

Findings. Decentralization substantially reduces effective blast radius, while broad information-sharing amplifies hidden-objective spread and emergent coordination even when only one agent is compromised.

Contribution. Connects scalable-oversight and multi-agent safety research to a design-space perspective on how organizational structure and information flow shape cascading risk in agentic systems.

Interaction-Level Governance for Healthcare LLMs

Interaction-level alignment · Lead author · Paper targeted for npj Digital Health

Framework. A three-layer governance framework that stratifies each interaction by clinical risk, user state, and decision proximity rather than treating all queries symmetrically. Risk tiers map to response strategies (clarification, escalation, bounded non-directive guidance), addressing the “gray-area” failure mode where models oscillate between sycophancy and blanket refusal.

Uncertainty mechanism. Proposed inter-model agreement on clinical risk assessment across multiple LLMs as an uncertainty signal for conservative arbitration.

Ongoing. Investigating RLHF and preference-optimization approaches to operationalize interaction-level safety policies as post-training targets.

Honesty–Sycophancy Tradeoff in LLMs with Evolving User Preferences

Incentives and trust in agentic systems · Sole author · Working paper

A formal model of the strategic tension providers face between honesty (long-run trust) and sycophancy (short-run engagement) when users have limited ability to detect sycophancy and preferences drift with model output. Proposes *trust* as a benchmark metric for alignment and governance evaluation, and identifies conditions under which honesty is the equilibrium strategy.

PUBLICATIONS AND RESEARCH (ECONOMICS OF DIGITAL PLATFORMS)

Gal-Or, E., Gal-Or, R., & Penmetsa, N. (2018). “The Role of User Privacy Concerns in Shaping Competition between Platforms.” *Information Systems Research*.

Gal-Or, E., Gal-Or, R., & Penmetsa, N. (2018). “Can platform competition support market segmentation? Network externalities versus matching efficiency in equity crowdfunding markets.” *Journal of Economics and Management Strategy*.

Gal-Or, E., May, J. H., & Penmetsa, N. (2015). “Dynamic Pricing of New Services in Subscription Markets.” *Production and Operations Management*.

Working Papers Relevant to AI Safety

Penmetsa & Anand. *Organization of Platform Markets: Coordination, Decision Rights and Information*. Revising for resubmission, *Management Science*. Formal analysis of centralized vs. decentralized coordination under agent competition and information asymmetry — directly extended in “Secret Loyalties.” [SSRN]

Penmetsa, Anand, & Goyal. *The Ethical Flywheel: Responsible Platforms and Thriving Markets*. Dynamic model of ethical vs. profit-maximizing platforms under bounded rationality; relevant to reputation systems, incentive design, and alignment of platform-mediated agents.

AI SAFETY TRAINING

BlueDot Impact — Technical AI Safety Course, 2026. RLHF, interpretability, evaluations, governance, frontier safety.

BlueDot Impact — AGI Strategy Course, 2026.

MLCommons — Member, AI Risk and AI Safety & Science Working Groups (2023–present).

TECHNICAL SKILLS

ML & Alignment. Inspect AI, multi-agent agentic evaluations, RLHF and RLAIIF, DPO, constitutional AI, prompt engineering, evaluation pipeline design.

Programming. Python (primary), scientific Python stack (NumPy, pandas), Jupyter, Git, API integration with frontier LLMs.

Modeling. Game theory, mechanism design, incentive analysis, formal modeling of multi-agent systems.

SELECTED RECENT TALKS (AI & DIGITAL PLATFORMS)

“Ethics and Platform Responsibility,” INFORMS Annual Meeting, Seattle, 2024.

“Ethics and Platforms,” POMS Annual Conference, Minneapolis, 2024.

“Organization of Platform Markets: Coordination, Decision Rights and Information,” Theory in Economics of Information Systems Workshop, San Francisco, 2023; INFORMS, Phoenix, 2023; CIST, Phoenix, 2023.

PROFESSIONAL SERVICE

Program Chair, Conference on Information Systems and Technology — *Amplified Intelligence: Economic, Work, and Societal Implications*, 2025. Reviewer for *Management Science*, *Information Systems Research*, *MSOM*, *JEMS*, *POM*, and *Decision Sciences*.

AWARDS

Rockwell Foundation Fellowship, University of Pittsburgh (2010–11). Doctoral Fellowship, University of Pittsburgh (2010–14). Multiple Excellence Awards, Citigroup (2008–09).